NATIONAL UNIVERSITY OF SINGAPORE

SEMESTER I, 2026/2027

MA1522 Linear Algebra for Computing **Graded Quiz 2**

1. Let $\boldsymbol{A}$ be a zero square matrix. Which of the following statements are true?

(i) $\boldsymbol{A}$ is a scalar matrix.

(ii) $\boldsymbol{A}$ is an upper triangular matrix.

(iii) $\boldsymbol{A}$ is a symmetric matrix.

Solution. Let $\boldsymbol{A} = (a_{ij})_{n\times n}$ be a square matrix of order n.

(i) A scalar matrix satisfies $a_{ij} = 0$ for $i \neq j$ and $a_{ii} = c$. If all entries of $\boldsymbol{A}$ are 0, $\boldsymbol{A}$ is a scalar matrix by setting $c = 0$.

(ii) An upper triangle matrix satisfies $a_{ij} = 0$ for $i > j$. If all entries of $\boldsymbol{A}$ are 0, $\boldsymbol{A}$ is an upper triangular matrix.

(iii) A symmetric matrix satisfies $a_{ij} = a_{ji}$ for all i, j. If all entries of $\boldsymbol{A}$ are 0, we have $a_{ij} = a_{ji} = 0$; so $\boldsymbol{A}$ is a symmetric matrix.

We conclude that (i), (ii) and (iii) are all true.

2. Which of the following two statements are true?

(i) A square matrix is diagonal $\Leftrightarrow$ it is both upper triangular and lower triangular.

(ii) A square matrix is diagonal $\Leftrightarrow$ it is both symmetric and triangular.

Solution. (i) Let $\boldsymbol{A} = (a_{ij})_{n\times n}$ be a square matrix.

$$\begin{aligned}\boldsymbol{A} \text{ is diagonal} &\Leftrightarrow a_{ij} = 0 \text{ for all } i \neq j\\ &\Leftrightarrow a_{ij} = 0 \text{ for all } i > j \text{ and } a_{ij} = 0 \text{ for all } i < j\\ &\Leftrightarrow \boldsymbol{A} \text{ is upper triangular and lower triangular.}\end{aligned}$$

(ii) A diagonal matrix is clearly symmetric and triangular.
Suppose $\boldsymbol{A}$ is symmetric and upper triangular. Then $\boldsymbol{A} = \boldsymbol{A}^{\mathrm{T}}$ is also lower triangular. It follows that $\boldsymbol{A}$ is a diagonal matrix.
Similarly, if $\boldsymbol{A}$ is symmetric and lower triangular, then $\boldsymbol{A}$ is a diagonal matrix.

We conclude that both (i) and (ii) are true.

3. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be square matrices of the same order. Which of the following statements are true?

(i) If both $\boldsymbol{A}$ and $\boldsymbol{B}$ are upper triangular, then $\boldsymbol{AB}$ is upper triangular.

(ii) If $\boldsymbol{A}$ is upper triangular and $\boldsymbol{B}$ is lower triangular, then $\boldsymbol{AB}$ is a diagonal matrix.

Solution. (i) Let $\boldsymbol{A} = (a_{ij})_{n\times n}$ and $\boldsymbol{B} = (b_{ij})_{n\times n}$. Then $a_{ij} = b_{ij} = 0$ for $i > j$.

Let $i > j$, the (i, j)-entry of $\boldsymbol{AB}$ is given by

$$(a_{i1}b_{1j} + \cdots + a_{ij}b_{jj}) + (a_{i,j+1}b_{j+1,j} + \cdots + a_{in}b_{nj}).$$

Since $a_{i1} = \cdots = a_{ij} = 0$, the terms in the first brackets is 0.

Since $b_{j+1,j} = \cdots = b_{n,j} = 0$, the terms in the second brackets is 0.

Hence, the (i, j)-entry of $\boldsymbol{AB}$ is 0. It follows that $\boldsymbol{AB}$ is upper triangular.

(ii) Let $\boldsymbol{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\boldsymbol{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. Then $\boldsymbol{A}$ is upper triangular and $\boldsymbol{B}$ is lower triangular.

However, $\boldsymbol{AB} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ is not a diagonal matrix.

We conclude that only (i) is true.

4. The matrix equation $\boldsymbol{Ax} = \boldsymbol{b}$ has the following general solutions: $\boldsymbol{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} + t\begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix}$, $t \in \mathbb{R}$.

If $\boldsymbol{A}$ has 3 rows and is in reduced row-echelon form, what can $\boldsymbol{A}$ and $\boldsymbol{b}$ be?

Solution.

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} + t\begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1+t \\ 2-t \\ 3+t \\ 4 \end{pmatrix}.$$

From third row/equation, we get $t = x_3 - 3$, and substitute into the first two rows/equations, we have the following system of linear equations:

$$\begin{cases} x_1 & & - & x_3 & & = & -2 \\ & x_2 & + & x_3 & & = & 5 \\ & & & & x_4 & = & 4 \end{cases}$$

with augmented matrix

$$\left(\begin{array}{cccc|c} 1 & 0 & -1 & 0 & -2 \\ 0 & 1 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right)$$

which is in reduced row-echelon form. All other systems of linear equations in reduced row-echelon forms giving the same general solutions are

$$\left(\begin{array}{c|c} \boldsymbol{A} & \boldsymbol{b} \end{array}\right) = \left(\begin{array}{cccc|c} 1 & 0 & -1 & 0 & -2 \\ 0 & 1 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

5. Let $\boldsymbol{A}$ be an $m \times n$ matrix, and $\boldsymbol{B}$ be an $n \times p$ matrix. Which of the following statements are true?

(i) If $\boldsymbol{A}$ has a zero column, then $\boldsymbol{AB}$ has a zero column.

(ii) If $\boldsymbol{B}$ has a zero column, then $\boldsymbol{AB}$ has a zero column.

Solution. (i) Let $\boldsymbol{A} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ and $\boldsymbol{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. Then $\boldsymbol{AB} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ has no zero column.

(ii) Let $\boldsymbol{b}_j$ be the jth column of $\boldsymbol{B}$. Then $\boldsymbol{Ab}_j$ is the jth column of $\boldsymbol{AB}$.
If the jth column of $\boldsymbol{B}$ is zero, i.e., $\boldsymbol{b}_j = \boldsymbol{0}$, then $\boldsymbol{Ab}_j = \boldsymbol{0}$; so the jth column of $\boldsymbol{AB}$ is $\boldsymbol{0}$.

We conclude that only (ii) is true.

6. Let $\boldsymbol{A}$ be a singular matrix. Which of the following two statements are true?

(i) There exists a nonzero square matrix $\boldsymbol{B}$ such that $\boldsymbol{AB} = \boldsymbol{0}$.

(ii) There exists a nonzero square matrix $\boldsymbol{B}$ such that $\boldsymbol{BA} = \boldsymbol{0}$.

Solution. (i) Since $\boldsymbol{A}$ is singular, the system $\boldsymbol{Ax} = \boldsymbol{0}$ has a solution $\boldsymbol{u} \neq \boldsymbol{0}$.
Let $\boldsymbol{B} = \begin{pmatrix} \boldsymbol{u} & \cdots & \boldsymbol{u} \end{pmatrix}$ be a square matrix whose columns are all $\boldsymbol{u}$. Then

$$\boldsymbol{AB} = \begin{pmatrix} \boldsymbol{Au} & \cdots & \boldsymbol{Au} \end{pmatrix} = \begin{pmatrix} \boldsymbol{0} & \cdots & \boldsymbol{0} \end{pmatrix} = \boldsymbol{0}.$$

(ii) If $\boldsymbol{A}$ is singular, then $\boldsymbol{A}^{\mathrm{T}}$ is singular. By (i) there exists a square matrix $\boldsymbol{C}$ such that $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{C} = \boldsymbol{0}$. Let $\boldsymbol{B} = \boldsymbol{C}^{\mathrm{T}}$. Then

$$(\boldsymbol{BA})^{\mathrm{T}} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{B}^{\mathrm{T}} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{C} = \boldsymbol{0}.$$

So $\boldsymbol{BA} = \boldsymbol{0}^{\mathrm{T}} = \boldsymbol{0}$.

We conclude that both (i) and (ii) are true.

7. Let $\boldsymbol{A}$ be an $m \times n$ matrix and $\boldsymbol{B}$ be an $n \times m$ matrix such that $\boldsymbol{AB} = \boldsymbol{I}_m$. Find the relation between m and n.

Solution. Let $\boldsymbol{R}$ be a row-echelon form of $\boldsymbol{A}$. Then there exist elementary matrices $\boldsymbol{E}_1, \ldots, \boldsymbol{E}_k$ such that $\boldsymbol{R} = \boldsymbol{E}_k \cdots \boldsymbol{E}_1 \boldsymbol{A}$. So

$$\boldsymbol{RB} = \boldsymbol{E}_k \cdots \boldsymbol{E}_1.$$

Similar to Question 5(ii), if the last row of $\boldsymbol{R}$ is zero, then the last row of $\boldsymbol{E}_k \cdots \boldsymbol{E}_1$ is zero. This leads to a contradiction because $\boldsymbol{E}_k \cdots \boldsymbol{E}_1$ is invertible.

Hence, all rows of $\boldsymbol{R}$ are nonzero. Note that $\boldsymbol{R}$ is also an $m \times n$ matrix. So

$$\begin{aligned} m &= \text{the number of rows of } \boldsymbol{R} \\ &= \text{the number of nonzero rows of } \boldsymbol{R} \\ &= \text{the number of pivot columns of } \boldsymbol{R} \\ &\leq \text{the number of columns of } \boldsymbol{R} \\ &= n. \end{aligned}$$

8. Let $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ be square matrices of the same order, in which $\boldsymbol{A}$ is invertible. Which of the following statements are true?

(i) If $\boldsymbol{AB} = \boldsymbol{AC}$, then $\boldsymbol{B} = \boldsymbol{C}$.
(ii) If $\boldsymbol{BA} = \boldsymbol{CA}$, then $\boldsymbol{B} = \boldsymbol{C}$.
(iii) If $\boldsymbol{AB} = \boldsymbol{CA}$, then $\boldsymbol{B} = \boldsymbol{C}$.

Solution. (i) $\boldsymbol{AB} = \boldsymbol{AC} \Rightarrow \boldsymbol{A}^{-1}\boldsymbol{AB} = \boldsymbol{A}^{-1}\boldsymbol{AC} \Rightarrow \boldsymbol{IB} = \boldsymbol{IC} \Rightarrow \boldsymbol{B} = \boldsymbol{C}$.
(ii) $\boldsymbol{BA} = \boldsymbol{CA} \Rightarrow \boldsymbol{BAA}^{-1} = \boldsymbol{CAA}^{-1} \Rightarrow \boldsymbol{BI} = \boldsymbol{CI} \Rightarrow \boldsymbol{B} = \boldsymbol{C}$.
(iii) Let $\boldsymbol{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Then $\boldsymbol{A}$ is invertible with $\boldsymbol{A}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$.
Let $\boldsymbol{B} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $\boldsymbol{C} = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$. Then $\boldsymbol{AB} = \boldsymbol{CA} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.
We conclude that only (i) and (ii) are true.

9. Let $\boldsymbol{A}_1, \boldsymbol{A}_2, \ldots, \boldsymbol{A}_n$ be square matrices of the same order such that $\boldsymbol{A}_1\boldsymbol{A}_2 \cdots \boldsymbol{A}_n = \boldsymbol{I}$. Which of the following statements are true?

(i) $\boldsymbol{A}_n\boldsymbol{A}_{n-1} \cdots \boldsymbol{A}_1 = \boldsymbol{I}$.
(ii) $\boldsymbol{A}_2\boldsymbol{A}_3 \cdots \boldsymbol{A}_n\boldsymbol{A}_1 = \boldsymbol{I}$.

Solution. (i) Let $\boldsymbol{A}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\boldsymbol{A}_2 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$, $\boldsymbol{A}_3 = \begin{pmatrix} 0 & \frac{1}{2} \\ 1 & 0 \end{pmatrix}$. Then

$$\boldsymbol{A}_1\boldsymbol{A}_2\boldsymbol{A}_3 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \boldsymbol{A}_3\boldsymbol{A}_2\boldsymbol{A}_1 = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix}.$$

(ii) By assumption, $\boldsymbol{A}_1^{-1} = \boldsymbol{A}_2 \cdots \boldsymbol{A}_n$; so $\boldsymbol{A}_2 \cdots \boldsymbol{A}_n\boldsymbol{A}_1 = \boldsymbol{I}$.
We conclude that only (ii) is true.

10. Which of the following statements are true?

(i) $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.

(ii) $\det\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aei + bfg + cdh - afh - bdi - ceg.$

(iii) $\det\begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} = afkp + bglm + chin + dejo - ahkn - belo - cfip - dgjm.$

Solution. Diagonal expansion for determinant is only valid for square matrices of order 2 or 3. So only (i) and (ii) are true.